pyAmpliCol at one loop

Benchmarks of the recursive one-loop engine against MadLoop, Recola and OpenLoops

report 2026-11-08 · newest engine measured `ecc80899` · finite part, binary64, one pinned thread, $B = 128$
Table 1: engine `ca602436`, G1 2026-11-06, AMD EPYC 9754, shared host · Table 2: engine `ecc80899`



Gate pass ×0.25 · fail ×1.4 · open ×1.0 · <5 ms: not gated ×1.4 — time ÷ limit
Stages inputs · numerator · scatter · NinjaRS · masters · glue
missing (why) · no colour: generation, LC · † within estimate · ¶ reference's own (DD) · ‡ user decision
ML MadLoop · Rc Recola · OL OpenLoops

Table 1: **Born-interfered one-loop processes** (the one-loop amplitude interfered with the Born). How to read it: at the end.

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	pyAmpliCol: the most recent measured configuration — `loop` · $2 \to 2$ to $2 \to 5$: engine `ca602436`, native `loop-native-v2-slice-ca602436 (4cbb549f…)`, G1 2026-11-06 · AMD EPYC 9754, shared host			
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 2$	$gg \to gg$ 4 legs · QCD · 24 families (LC 12) Rc·OL·ML ≤ 1.3e−12 16 pts · median 1.1e−14 · †	**5.70 s** ×40 0.57 GiB peak · plan 0.0240 MB **Rc 142 ms** · ML 9.70 s	**1.29 ms** ×0.64 $B = 128$ **ML 2.02 ms** · OL 3.20 ms · Rc 10.0 ms no gate (2→2) numerator 51 %	**0.596 ms** LC/FC 0.46 $B = 128$ · FC 1.29 ms
$2 \to 3$	$u\bar{u} \to d\bar{d}g$ 5 legs · QCD · 42 families (LC 27) ML·OL ≤ 7.9e−13 · Rc 3.3e−10 16 pts · median 4.8e−14 · ¶	**6.73 s** ×17 0.58 GiB peak · plan 0.0300 MB **Rc 406 ms** · ML 9.87 s	**1.72 ms** ×1.1 $B = 128$ **OL 1.52 ms** · ML 3.17 ms · Rc 3.18 ms <5 ms: not gated ×0.87 numerator 38 %	**1.12 ms** LC/FC 0.65 $B = 128$ · FC 1.72 ms
	$u\bar{u} \to u\bar{u}g$ 5 legs · QCD · 68 families (LC 42) OL·ML ≤ 1.2e−11 · Rc 7.2e−11 16 pts · median 1.4e−13 · ¶	**7.07 s** ×15 0.58 GiB peak · plan 0.0360 MB **Rc 462 ms** · ML 10.9 s	**2.64 ms** ×0.90 $B = 128$ **OL 2.92 ms** · Rc 5.55 ms · ML 8.59 ms <5 ms: not gated ×0.69 numerator 42 %	**1.59 ms** LC/FC 0.60 $B = 128$ · FC 2.64 ms
	$u\bar{u} \to e^-e^+g$ 5 legs · QCD · 11 families (LC 7) ML·Rc ≤ 7.5e−14 16 pts · median 6.8e−15	**10.1 s** ×14 0.61 GiB peak · plan 0.0530 MB **Rc 701 ms** · ML 14.0 s	**0.788 ms** ×0.83 $B = 128$ **ML 0.948 ms** · Rc 1.08 ms · OL no library <5 ms: not gated ×0.83 numerator 36 %	**0.446 ms** LC/FC 0.57 $B = 128$ · FC 0.788 ms
	$u\bar{u} \to ggg$ 5 legs · QCD · 90 families (LC 62) ML·Rc ≤ 1.5e−12 · OL 8.4e−11 16 pts · median 4.9e−14 · ¶	**6.62 s** ×10 0.58 GiB peak · plan 0.0490 MB **Rc 634 ms** · ML 17.6 s	**4.34 ms** ×0.35 $B = 128$ **OL 12.4 ms** · Rc 25.3 ms · ML 28.2 ms <5 ms: not gated ×0.27 numerator 53 %	**2.38 ms** LC/FC 0.55 $B = 128$ · FC 4.34 ms
	$gg \to ggg$ 5 legs · QCD · 154 families (LC 77) OL·Rc·ML ≤ 3.7e−7 16 pts · median 5.8e−13 · †	**8.10 s** ×1.2 0.59 GiB peak · plan 0.112 MB **Rc 6.63 s** · ML 47.0 s	**20.1 ms** ×0.21 $B = 128$ **OL 97.0 ms** · ML 247 ms · Rc 351 ms pass ×0.16 numerator 50 %	**7.59 ms** LC/FC 0.38 $B = 128$ · FC 20.1 ms
$2 \to 4$	$u\bar{u} \to e^-e^+gg$ 6 legs · QCD · 69 families (LC 33) Rc·ML ≤ 2.2e−10 16 pts · median 6.0e−14 · †	**9.53 s** ×11 0.62 GiB peak · plan 0.0680 MB **Rc 833 ms** · ML 24.8 s	**4.21 ms** ×0.41 $B = 128$ **Rc 10.2 ms** · ML 27.3 ms · OL no library <5 ms: not gated ×0.32 numerator 54 %	**1.71 ms** LC/FC 0.41 $B = 128$ · FC 4.21 ms
	$gg \to t\bar{t}gg$ 6 legs · QCD · 1 086 families (LC 375) OL·Rc·ML ≤ 8.8e−9 16 pts · median 1.2e−11 · †	**9.98 s** ×0.16 0.64 GiB peak · plan 0.320 MB **Rc 61.4 s** · ML 218 s	**79.8 ms** ×0.17 $B = 128$ **OL 472 ms** · Rc 1.97 s · ML 5.45 s pass ×0.13 numerator 44 %	**28.2 ms** LC/FC 0.35 $B = 128$ · FC 79.8 ms
	$u\bar{u} \to gggg$ 6 legs · QCD · 628 families (LC 375) Rc 8.0e−12 · OL 2.3e−10 16 pts · median 1.7e−12 · †	**9.86 s** ×0.30 0.65 GiB peak · plan 0.382 MB **Rc 32.8 s**	**65.3 ms** ×0.17 $B = 128$ **OL 374 ms** · Rc 908 ms · ML not exported pass ×0.13 numerator 57 %	**25.3 ms** LC/FC 0.39 $B = 128$ · FC 65.3 ms
	$gg \to gggg$ 6 legs · QCD · 1 100 families (LC 550) ML·Rc·OL ≤ 1.5e−7 8 pts · median 2.5e−11 · †	**14.0 s** ×0.0066 0.89 GiB peak · plan 1.23 MB **Rc 35.4 min** · ML 144.6 min	**307 ms** ×0.11 $B = 128$ **OL 2.84 s** · Rc 17.6 s · ML 38.7 s pass ×0.083 numerator 51 %	**126 ms** LC/FC 0.41 $B = 128$ · FC 307 ms

pyAmpliCol: the most recent measured configuration — loop · $2 \to 2$ to $2 \to 5$: engine ca602436, native loop-native-v2-slice-ca602436 (4cbb549f…), G1 2026-11-06 · AMD EPYC 9754, shared host

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 5$	$u\bar{u} \to e^-e^+d\bar{d}g$ 7 legs · QCD · 242 families (LC 114) OL·ML·Rc ≤ 4.5e−11 16 pts · median 5.3e−13	**12.1 s** ×1.8 0.63 GiB peak · plan 0.0960 MB **Rc 6.73 s** · ML 58.6 s	**13.9 ms** ×0.39 **Rc 35.3 ms** · OL 51.0 ms · ML 132 ms $B = 128$ pass ×0.30 numerator 48 %	**5.75 ms** LC/FC 0.41 $B = 128$ · FC 13.9 ms
	$u\bar{u} \to e^-e^+ggg$ 7 legs · QCD · 463 families (LC 164) ML·Rc·OL ≤ 4.3e−11 16 pts · median 1.4e−12 · †	**11.1 s** ×0.83 0.64 GiB peak · plan 0.179 MB **Rc 13.4 s** · ML 146 s	**47.9 ms** ×0.19 **Rc 251 ms** · OL 346 ms · ML 1.42 s $B = 128$ pass ×0.15 numerator 63 %	**9.42 ms** LC/FC 0.20 $B = 128$ · FC 47.9 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · QCD · 579 families (LC 164) OL·Rc·ML ≤ 1.1e−8 16 pts · median 2.7e−11 · †	**11.8 s** ×0.43 0.64 GiB peak · plan 0.176 MB **Rc 27.7 s** · ML 165 s	**51.1 ms** ×0.13 **OL 404 ms** · Rc 620 ms · ML 3.04 s $B = 128$ pass ×0.097 numerator 57 %	**10.1 ms** LC/FC 0.20 $B = 128$ · FC 51.1 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · EW · 1 599 families OL 2.7e−10 · ML 2.8e−9 16 pts · median 4.5e−10	**276 s** ×12 9.24 GiB peak · plan 0.445 MB **Rc 23.5 s** · ML 11.9 min	**131 ms** ×0.13 **Rc 987 ms** · OL 1.40 s · ML 7.07 s $B = 128$ pass ×0.10 numerator 46 % model: the workspace's py3 G_mu copy	***not supported*** refused by design: EW LC/NLC is never supported (user 2026-10-30, Q4; typed colour-declaration-coverage)
	$u\bar{u} \to ggggg$ 7 legs · QCD · 5 258 families (LC 2 712) ML 2.1e−12 · Rc 2.9e−10 3 pts · median 7.0e−11 · ¶	**88.2 s** ×0.0086 2.76 GiB peak · plan 14.7 MB **Rc 170.3 min** · ML 12.2 h	**1.31 s** ≥×0.0072 **Rc ≤182 s** · ML 569 s · OL no library $B = 128$ pass ×0.0023 numerator 62 %	**438 ms** LC/FC 0.34 $B = 128$ · FC 1.31 s
	$gg \to ggggg$ 7 legs · QCD · 4 707 families (LC 4 707) NJet DD 1.5e−7 10 pts · median 3.4e−9 · † · top-free (particle_veto = [t]), Nf = 5 ‡ pts 4, 8 above 2.0e−8: accepted (user 2026-10-25)	**133 s** 4.39 GiB peak · plan 20.2 MB ML diagrams 17.4 h, stopped in export · Rc unfinished	**4.99 s** $B = 128$ no gate (no matched provider) numerator 52 %	**2.68 s** LC/FC 0.54 $B = 128$ · FC 4.99 s

$2 \to 6$: pyAmpliCol only, no provider exists at $2 \to 6$, so no ratio, no gate and no colour · engine 2f3f76d9 (work/v2-mf-8g-3), native loop-native-mf-2to6-2f3f76d9-lto (2bae17f9…), an ancestor of the engine above, not re-run on it · shared host, indicative · internal checks only

$2 \to 6$	$gg \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 1.0e−11 · E ≤ 1.4e−3 (2 pts above 10^{-5}) · binary64 = DD to 7.4e−5 · rescue 11/16 · NGluon DD to 6.3e−13 · **FC** ε^{-2} ≤ 8.4e−11 · E ≤ 5.1e−3 (2 pts above 10^{-5}) · Catani ε^{-1} ≤ 7.1e−5 · binary64 = DD to 1.2e−4 · rescue 11/16 Ward FC not run, Ward LC not asserted	**55.3 min** FC 74.0 GiB peak · plan 486 MB · store 25.8 min artifact 0.53 GB · load 4.21 s (1.06 GiB) **LC** 46.1 min · 63.9 GiB · artifact 0.364 GB	**201 s** 1 block of 16 $B = 16$ no provider: no gate inputs 55 % peak 40.1 GiB	**98.0 s** 1 block of 128 $B = 128$ numerator 54 % peak 27.2 GiB
	$u\bar{u} \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 4.6e−12 · E ≤ 6.2e−7 · binary64 = DD to 8.4e−11 · rescue 1/16 · Ward ≤ 6.5e−12 · **FC** ε^{-2} ≤ 1.1e−11 · E ≤ 5.9e−6 · Catani ε^{-1} ≤ 3.3e−8 · binary64 = DD to 1.7e−10 · rescue 2/16 · Ward ≤ 2.1e−11	**73.6 min** FC 92.9 GiB peak · plan 814 MB · store 449 s artifact 0.143 GB · load 3.85 s (1.66 GiB) **LC** 20.5 min · 47.0 GiB · artifact 0.047 GB	**62.1 s** 1 block of 128 $B = 128$ no provider: no gate numerator 77 % peak 31.7 GiB	**10.9 s** 1 block of 128 $B = 128$ numerator 60 % peak 12.6 GiB
	$u\bar{u} \to u\bar{u}gggg$ 8 legs · QCD · 40 454 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 6.7e−14 · E ≤ 7.5e−7 · binary64 = DD to 1.3e−11 · rescue 2/16 · Ward ≤ 4.3e−12 · **FC** ε^{-2} ≤ 2.0e−13 · E ≤ 6.5e−6 · Catani ε^{-1} ≤ 6.6e−11 · binary64 = DD to 1.3e−11 · rescue 4/16 · Ward ≤ 8.5e−12 (DD 1.1e−28)	**29.8 min** FC 42.5 GiB peak · plan 315 MB · store 46.4 s artifact 0.038 GB · load 1.51 s (0.68 GiB) **LC** 506 s · 18.2 GiB · artifact 0.009 GB	**19.2 s** 1 block of 128 $B = 128$ no provider: no gate numerator 75 % peak 36.0 GiB	**2.67 s** 1 block of 128 $B = 128$ numerator 55 % peak 7.38 GiB

Table 2: **Loop-induced processes** ($|A_{\text{loop}}|^2$; the equivalent of Table 1), timed with the unchecked reduction setting[1]. How to read it: at the end.

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked (re-timed rows)
pyAmpliCol: engine e852f448, native loop-native-v2-slice-e852f448 (c3fda0c4…), engine ecc80899, native loop-native-lit-unchecked-b308254d (0d26af4a…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells of the 17 re-timed rows: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside; the other rows keep their earlier record (auto, checked); gate on TIRust · measured 2026-11-09 · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine ecc80899 (before lit-dress) replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$2 \to 1 \cdot 1 \to 2$		
$gg \to h$ 3 legs · QCD · LI-A1 AN 2.6e−13 · OL 6.2e−13 · ML 9.2e−13 · Rc 9.3e−13 · pass · 6 pts	**2.96 s** validation probes 3.10 s (double-double) 0.56 GiB peak · plan 0.0100 MB	**0.203 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→1) **NinjaRS per amplitude** (auto) · TIRust 0.192 ms (128/128) earlier record: checked backends, not re-timed unchecked
$h \to gg$ 3 legs · QCD · LI-A2 AN 6.2e−14 · Rc 6.3e−13 · ML 6.3e−13 · pass · 3 pts	**2.95 s** validation probes 3.08 s (double-double) 0.56 GiB peak · plan 0.0100 MB	**0.216 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→2) **NinjaRS per amplitude** (auto) · TIRust 0.205 ms (128/128) earlier record: checked backends, not re-timed unchecked
$h \to aa$ 3 legs · QED · LI-A3, LI-E1 AN 2.9e−14 · ML 1.9e−13 · Rc 2.0e−13 · pass · 3 pts	**72.1 s** validation probes 73.7 s (double-double) 3.23 GiB peak · plan 0.0270 MB	**0.639 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→2) **NinjaRS per amplitude** (auto) · TIRust 0.620 ms (128/128) earlier record: checked backends, not re-timed unchecked
$h \to za$ 3 legs · QED · LI-A4, LI-E1 ML 1.6e−14 · AN 1.7e−14 · Rc 7.9e−14 · pass · 3 pts	**75.0 s** validation probes 179 s (double-double) 4.49 GiB peak · plan 0.0350 MB	**0.786 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→2) **NinjaRS per amplitude** (auto) · TIRust 0.684 ms (128/128) earlier record: checked backends, not re-timed unchecked
$aa \to h$ 3 legs · QED · LI-E2 AN 5.2e−14 · ML 2.2e−13 · Rc 2.2e−13 · pass · 3 pts	**72.4 s** validation probes 73.6 s (double-double) 3.25 GiB peak · plan 0.0270 MB	**0.614 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→1) **NinjaRS per amplitude** (auto) · TIRust 0.598 ms (128/128) earlier record: checked backends, not re-timed unchecked
$2 \to 2$		
$gg \to aa$ 4 legs · QCD · LI-A5 ML 3.4e−14 · Rc 4.5e−14 · AN 4.6e−14 · OL 4.7e−14 · pass · 6 pts	**4.06 s** validation probes 6.94 s (double-double) 0.67 GiB peak · plan 0.0210 MB	**3.42 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 2.17 ms (128/128) earlier record: checked backends, not re-timed unchecked
$gg \to zz$ 4 legs · QCD · LI-21 Rc 4.6e−12 · OL 5.9e−12 · ML 6.6e−12 · pass · 3 pts	**4.56 s** validation probes 9.06 s (double-double) 0.70 GiB peak · plan 0.0290 MB	**8.61 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 5.43 ms (128/128) earlier record: checked backends, not re-timed unchecked
$gg \to w^+w^-$ 4 legs · QCD · LI-22 OL 1.7e−13 · Rc 5.7e−13 · ML 7.8e−13 · pass · 3 pts	**5.00 s** validation probes 10.4 s (double-double) 0.71 GiB peak · plan 0.0420 MB	**9.55 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 4.53 ms (128/128) earlier record: checked backends, not re-timed unchecked
$gg \to za$ 4 legs · QCD · LI-23 Rc 1.7e−14 · OL 1.9e−14 · ML 4.2e−14 · pass · 3 pts	**4.84 s** validation probes 8.54 s (double-double) 0.69 GiB peak · plan 0.0360 MB	**4.88 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 2.79 ms (128/128) earlier record: checked backends, not re-timed unchecked
$gg \to hh$ 4 legs · QCD · LI-24, LI-A7 OL 3.2e−14 · Rc 9.3e−14 · ML 1.6e−13 · pass · 3 pts	**3.42 s** ×35 validation probes 5.09 s (double-double) 0.60 GiB peak · plan 0.0100 MB Rc 98.4 ms	**0.437 ms** ×1.3 **ML 0.331 ms** · Rc 0.368 ms · OL 0.625 ms $B = 128$ · 3 runtimes ×1.0 no gate (2→2) **TIRust** (tirust) glue 41 % vs 256-bit truth (128 pts): max 3.4e−9 **NinjaRS per amplitude** (ninjars) 0.612 ms ×1.9 NinjaRS 34 % vs 256-bit truth (128 pts): max 8.8e−11

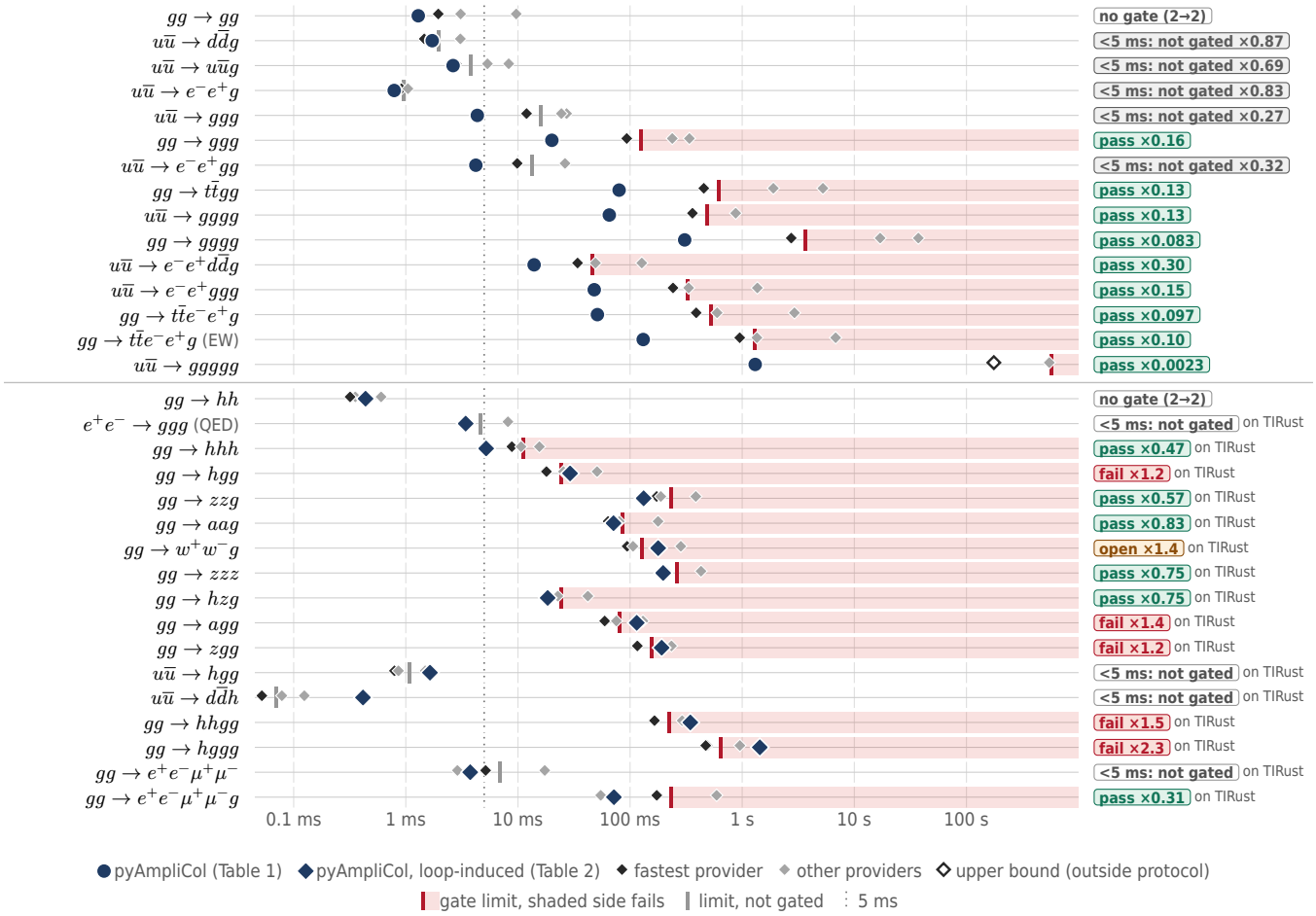
[1]The setting is off by default. A default call (reduction="auto", checks of the backends on) runs TIRust at the target accuracy (10^{-5}) with the per-point NinjaRS fallback; it does not get these times. Its served fractions and checked times are lit-perf's records in data/li-timing.json.

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked (re-timed rows)
pyAmpliCol: engine e852f448, native loop-native-v2-slice-e852f448 (c3fda0c4…), engine ecc80899, native loop-native-lit-unchecked-b308254d (0d26af4a…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells of the 17 re-timed rows: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside; the other rows keep their earlier record (auto, checked); gate on TIRust · measured 2026-11-09 · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine ecc80899 (before lit-dress) replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$2 \to 2$		
$gg \to hz$ 4 legs · QCD · LI-25 ML 5.5e−14 · Rc 6.0e−14 · OL 2.0e−13 · pass · 3 pts	**5.49 s** validation probes 6.12 s (double-double) 0.68 GiB peak · plan 0.0280 MB	**1.28 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 0.956 ms (128/128) earlier record: checked backends, not re-timed unchecked
$gg \to hg$ 4 legs · QCD · LI-26, LI-A7 OL 5.3e−15 · ML 7.1e−15 · Rc 1.3e−14 · pass · 3 pts	**3.30 s** validation probes 4.45 s (double-double) 0.60 GiB peak · plan 0.0110 MB	**1.45 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 0.908 ms (128/128) earlier record: checked backends, not re-timed unchecked
$u\bar{u} \to hg$ 4 legs · QCD · LI-26, LI-A7 Rc 2.0e−15 · ML 3.9e−15 · OL 5.1e−15 · pass · 3 pts	**3.34 s** validation probes 3.51 s (double-double) 0.57 GiB peak · plan 0.0100 MB	**0.264 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 0.248 ms (128/128) earlier record: checked backends, not re-timed unchecked
$ug \to hu$ 4 legs · QCD · LI-26 OL 5.0e−15 · ML 5.3e−15 · Rc 9.4e−15 · pass · 3 pts	**3.48 s** validation probes 3.52 s (double-double) 0.57 GiB peak · plan 0.0100 MB	**0.258 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 0.249 ms (128/128) earlier record: checked backends, not re-timed unchecked
$gg \to zg$ 4 legs · QCD · LI-27 Rc 4.3e−14 · OL 1.1e−13 · ML 1.9e−13 · pass · 3 pts	**4.83 s** validation probes 11.6 s (double-double) 0.70 GiB peak · plan 0.0290 MB	**13.2 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 5.90 ms (128/128) earlier record: checked backends, not re-timed unchecked
$gg \to ag$ 4 legs · QCD · LI-27 OL 1.6e−13 · ML 1.9e−13 · Rc 1.9e−13 · pass · 3 pts	**4.59 s** validation probes 10.4 s (double-double) 0.69 GiB peak · plan 0.0230 MB	**6.88 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 3.43 ms (128/128) earlier record: checked backends, not re-timed unchecked
$e^+e^- \to ha$ 4 legs · QED · LI-A4, LI-E3 Rc 1.2e−14 · ML 1.3e−14 · pass · 3 pts	**120 s** validation probes 127 s (double-double) 5.65 GiB peak · plan 0.0670 MB	**1.14 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 1.22 ms (128/128) earlier record: checked backends, not re-timed unchecked
$aa \to aa$ 4 legs · QED · LI-A6, LI-E2 Rc 1.5e−14 · ML 1.5e−14 · AN 2.0e−14 · pass · 3 pts	**112 s** validation probes 115 s (double-double) 6.11 GiB peak · plan 0.0390 MB	**7.50 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 4.05 ms (128/128) earlier record: checked backends, not re-timed unchecked
$aa \to zz$ 4 legs · QED · LI-E2 ML 2.5e−14 · Rc 2.6e−14 · pass · 3 pts	**127 s** validation probes 305 s (double-double) 6.83 GiB peak · plan 0.0920 MB	**18.7 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 10.2 ms (128/128) earlier record: checked backends, not re-timed unchecked
$aa \to hh$ 4 legs · QED · LI-E2 Rc 6.1e−13 · ML 6.1e−13 · pass · 3 pts	**88.8 s** validation probes 97.4 s (double-double) 3.66 GiB peak · plan 0.0340 MB	**1.72 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 1.33 ms (128/128) earlier record: checked backends, not re-timed unchecked
$e^+e^- \to hh$ 4 legs · QED · LI-E3 Rc 1.3e−10 · ML 1.3e−10 · pass · 3 pts	**116 s** validation probes 123 s (double-double) 5.63 GiB peak · plan 0.0390 MB	**0.988 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 1.03 ms (128/128) earlier record: checked backends, not re-timed unchecked
$1 \to 3$		
$h \to ggg$ 4 legs · QCD · LI-37 Rc 7.9e−13 · ML 7.0e−12 · partial · 3 pts	**3.31 s** validation probes 4.59 s (double-double) 0.60 GiB peak · plan 0.0110 MB	**1.58 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→3) **NinjaRS per amplitude** (auto) · TIRust 1.03 ms (128/128) earlier record: checked backends, not re-timed unchecked

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked (re-timed rows)
pyAmpliCol: engine e852f448, native loop-native-v2-slice-e852f448 (c3fda0c4...), engine ecc80899, native loop-native-lit-unchecked-b308254d (0d26af4a...) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells of the 17 re-timed rows: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside; the other rows keep their earlier record (auto, checked); gate on TIRust · measured 2026-11-09 · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine ecc80899 (before lit-dress) replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$1 \to 3$		
$z \to ggg$ 4 legs · QCD · LI-37 ML 7.8e−12 · Rc 3.2e−11 · partial · 3 pts	**4.78 s** validation probes 10.7 s (double-double) 0.70 GiB peak · plan 0.0290 MB	**9.99 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→3) **NinjaRS per amplitude** (auto) · TIRust: 42/128 failed closed, ruling (d) earlier record: checked backends, not re-timed unchecked
$2 \to 3$		
$e^+e^- \to ggg$ 5 legs · QED · LI-E4 ML 1.1e−10 · Rc 1.4e−8 · partial · 3 pts	**52.8 s** ×289 validation probes 58.4 s (double-double) 3.68 GiB peak · plan 0.0380 MB Rc 182 ms	**3.42 ms** ×0.96 **Rc 3.55 ms** · ML 8.45 ms · OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) $B = 128$ · 3 runtimes ×1.8 <5 ms: not gated on TIRust **TIRust** (tirust) numerator 56 % vs 256-bit truth (128 pts): max 3.5e−7 **NinjaRS per amplitude** (ninjars) 7.12 ms ×2.0 NinjaRS 44 % vs 256-bit truth (128 pts): max 4.6e−10
$gg \to hhh$ 5 legs · QCD · LI-31 ML 2.8e−13 · OL 3.2e−13 · Rc 8.1e−12 · pass · 3 pts	**5.38 s** ×47 validation probes 11.6 s (double-double) 0.68 GiB peak · plan 0.0120 MB Rc 114 ms	**5.20 ms** ×0.57 **Rc 9.16 ms** · ML 11.1 ms · OL 16.1 ms $B = 128$ · 3 runtimes ×1.0 pass ×0.47 on TIRust **TIRust** (tirust) TIR moments 55 % vs 256-bit truth (128 pts): max 7.6e−7 **NinjaRS per amplitude** (ninjars) 5.65 ms ×0.62 NinjaRS 48 % **accuracy: 2/128 pts above 1.0e−5, worst 4.5e64, finite values where the checked mode refuses**
$gg \to hgg$ 5 legs · QCD · LI-32, LI-A7 OL 2.7e−10 · ML 2.7e−10 · Rc 2.8e−10 · partial · 3 pts	**6.91 s** ×37 validation probes 38.1 s (double-double) 0.77 GiB peak · plan 0.0190 MB Rc 187 ms	**29.2 ms** ×1.6 **OL 18.6 ms** · Rc 26.4 ms · ML 52.7 ms $B = 128$ · 3 runtimes ×1.1 fail ×1.2 on TIRust **TIRust** (tirust) numerator 55 % vs 256-bit truth (128 pts): max 9.2e−10 **NinjaRS per amplitude** (ninjars) 59.3 ms ×3.2 NinjaRS 50 % vs 256-bit truth (128 pts): max 8.0e−10 auto with the setting: values bitwise TIRust's, 0 per-point fallbacks, time ×0.98
$gg \to zzg$ 5 legs · QCD · LI-33 ML 2.0e−8 · Rc 2.0e−8 · OL 8.1e−8 · partial · 3 pts	**6.72 s** ×6.3 validation probes 229 s (double-double) 1.17 GiB peak · plan 0.0380 MB Rc 1.07 s	**132 ms** ×0.74 **OL 179 ms** · Rc 195 ms · ML 400 ms $B = 128$ · 3 runtimes ×1.0 pass ×0.57 on TIRust **TIRust** (tirust) numerator 76 % **accuracy: 18/128 pts above 1.0e−5, worst 1.2e−2, all served by the gate-0 PV last resort** **NinjaRS per amplitude** (ninjars) 386 ms ×2.2 NinjaRS 64 % vs 256-bit truth (128 pts): max 5.0e−8
$gg \to aag$ 5 legs · QCD · LI-33 ML 3.3e−11 · Rc 3.3e−11 · OL 4.8e−11 · partial · 3 pts	**8.94 s** ×28 validation probes 169 s (double-double) 0.98 GiB peak · plan 0.0300 MB Rc 314 ms	**71.1 ms** ×1.1 **OL 66.1 ms** · Rc 83.2 ms · ML 184 ms $B = 128$ · 3 runtimes ×1.1 pass ×0.83 on TIRust **TIRust** (tirust) numerator 78 % vs 256-bit truth (128 pts): max 7.0e−6 **NinjaRS per amplitude** (ninjars) 233 ms ×3.5 NinjaRS 66 % **accuracy: 1/128 pts above 1.0e−5, worst 3.1e−3, also in the checked mode**
$gg \to w^+w^-g$ 5 legs · QCD · LI-33 ML 8.6e−10 · Rc 3.9e−9 · OL 4.3e−8 · partial · 3 pts	**10.2 s** ×6.8 validation probes 267 s (double-double) 1.21 GiB peak · plan 0.0580 MB Rc 1.51 s	**178 ms** ×1.8 **OL 97.7 ms** · Rc 110 ms · ML 294 ms $B = 128$ · 3 runtimes ×1.0 open ×1.4 on TIRust **TIRust** (tirust) numerator 59 % **accuracy: 24/128 pts above 1.0e−5, worst 1.4e−2, all served by the gate-0 PV last resort** **NinjaRS per amplitude** (ninjars) 540 ms ×5.5 NinjaRS 70 % vs 256-bit truth (128 pts): max 6.2e−8
$gg \to zzz$ 5 legs · QCD · LI-34 ML 9.4e−12 · OL 2.8e−11 · Rc 2.6e−10 · partial · 3 pts	**22.3 s** ×24 validation probes 201 s (double-double) 1.07 GiB peak · plan 0.0330 MB Rc 937 ms	**197 ms** ×0.98 **OL 202 ms** · Rc 216 ms · ML 446 ms $B = 128$ · 3 runtimes ×1.0 pass ×0.75 on TIRust **TIRust** (tirust) numerator 84 % vs 256-bit truth (128 pts): max 2.3e−8 **NinjaRS per amplitude** (ninjars) 18/128 failed closed, lane status Unstable NinjaRS 60 % **accuracy: 20 pts fail (Unstable), the call timed whole**

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked (re-timed rows)
pyAmpliCol: engine e852f448, native loop-native-v2-slice-e852f448 (c3fda0c4…), engine ecc80899, native loop-native-lit-unchecked-b308254d (0d26af4a…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells of the 17 re-timed rows: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside; the other rows keep their earlier record (auto, checked); gate on TIRust · measured 2026-11-09 · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine ecc80899 (before lit-dress) replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$2 \to 3$		
$gg \to hzg$ 5 legs · QCD · LI-35 ML 1.9e−13 · Rc 2.1e−13 · OL 2.5e−13 · pass · 3 pts	**5.43 s** ×22 validation probes 31.6 s (double-double) 0.82 GiB peak · plan 0.0330 MB Rc 246 ms	**18.4 ms** ×0.98 **Rc 18.8 ms** · OL 23.5 ms · ML 43.6 ms $B = 128$ · 3 runtimes ×1.0 pass ×0.75 on TIRust **TIRust** (tirust) numerator 76 % vs 256-bit truth (128 pts): max 8.2e−9 **NinjaRS per amplitude** (ninjars) 2/128 failed closed, lane status Unstable, lane status MasterFailed NinjaRS 51 % **accuracy: 2 pts fail (MasterFailed, Unstable), the call timed whole**
$gg \to agg$ 5 legs · QCD · LI-35 Rc 1.3e−11 · ML 2.1e−11 · OL 3.9e−11 · partial · 3 pts	**6.18 s** ×7.4 validation probes 203 s (double-double) 1.35 GiB peak · plan 0.0460 MB Rc 836 ms	**115 ms** ×1.9 **OL 61.6 ms** · Rc 78.4 ms · ML 135 ms $B = 128$ · 3 runtimes ×1.0 fail ×1.4 on TIRust **TIRust** (tirust) numerator 70 % vs 256-bit truth (128 pts): max 3.3e−6 **NinjaRS per amplitude** (ninjars) 268 ms ×4.3 NinjaRS 55 % **accuracy: 1/128 pts above 1.0e−5, worst 6.8e−4, also in the checked mode**
$gg \to zgg$ 5 legs · QCD · LI-35 Rc 2.9e−6 · ML 2.9e−6 · partial · 3 pts	**6.50 s** ×3.8 validation probes 193 s (double-double) 1.47 GiB peak · plan 0.0520 MB Rc 1.72 s	**191 ms** ×1.6 **Rc 121 ms** · ML 243 ms · OL unavailable: no OpenLoops ls library serves g g > z g g: the public channel database of OpenLoops 2.1.5 (proclib/channels_public.rinfo, 2026-11-06) has zgg/zggg (ppvj2) but no zgggg channel; row ML/RC only (PLAN §3.2) $B = 128$ · 3 runtimes ×1.0 fail ×1.2 on TIRust **TIRust** (tirust) numerator 65 % **accuracy: 3/128 pts above 1.0e−5, worst 3.1e−5, all served by the gate-0 PV last resort** **NinjaRS per amplitude** (ninjars) 418 ms ×3.5 NinjaRS 54 % vs 256-bit truth (128 pts): max 6.2e−6
$u\bar{u} \to hgg$ 5 legs · QCD · LI-36 Rc 7.7e−10 · ML 7.8e−10 · OL 7.8e−10 · partial · 3 pts	**3.87 s** ×4.0 validation probes 5.62 s (double-double) 0.61 GiB peak · plan 0.0110 MB Rc 960 ms	**1.64 ms** ×2.0 **OL 0.822 ms** · Rc 0.892 ms · ML 1.55 ms $B = 128$ · 3 runtimes ×1.6 <5 ms: not gated on TIRust **TIRust** (tirust) glue 36 % vs 256-bit truth (128 pts): max 3.7e−9 **NinjaRS per amplitude** (ninjars) 3.05 ms ×3.7 NinjaRS 34 % vs 256-bit truth (128 pts): max 3.5e−8
$u\bar{u} \to d\bar{d}h$ 5 legs · QCD · LI-36 OL 8.1e−11 · Rc 1.2e−10 · ML 1.2e−10 · pass · 3 pts	**4.23 s** ×14 validation probes 4.23 s (double-double) 0.58 GiB peak · plan 0.0100 MB Rc 293 ms	**0.415 ms** ×7.7 **Rc 0.0540 ms** · ML 0.0814 ms · OL 0.129 ms $B = 128$ · 3 runtimes ×4.2 <5 ms: not gated on TIRust **TIRust** (tirust) glue 90 % vs 256-bit truth (128 pts): max 4.1e−10 **NinjaRS per amplitude** (ninjars) 0.445 ms ×8.2 glue 85 % vs 256-bit truth (128 pts): max 4.3e−9
$2 \to 4$		
$gg \to hhgg$ 6 legs · QCD · LI-41 Rc 2.1e−9 · ML 2.8e−9 · OL 1.2e−7 · partial · 4 pts	**6.13 s** ×16 validation probes 512 s (double-double) 1.75 GiB peak · plan 0.0470 MB Rc 393 ms	**345 ms** ×2.0 **Rc 171 ms** · OL 302 ms · ML 365 ms $B = 128$ · 3 runtimes ×1.0 fail ×1.5 on TIRust **TIRust** (tirust) TIR moments 55 % vs 256-bit truth (32 pts): max 7.1e−7 **NinjaRS per amplitude** (ninjars) 789 ms ×4.6 NinjaRS 75 % vs 256-bit truth (32 pts): max 1.8e−6
$gg \to hggg$ 6 legs · QCD · LI-42 Rc 2.2e−11 · ML 3.0e−11 · OL 4.2e−11 · partial · 3 pts	**7.96 s** ×2.3 validation probes 14.6 min (double-double) 5.22 GiB peak · plan 0.100 MB Rc 3.46 s	**1.44 s** ×2.9 **Rc 491 ms** · OL 508 ms · ML 990 ms $B = 128$ · 3 runtimes ×1.1 fail ×2.3 on TIRust **TIRust** (tirust) numerator 47 % vs 256-bit truth (32 pts): max 5.9e−8 **NinjaRS per amplitude** (ninjars) 2.53 s ×5.2 NinjaRS 39 % **accuracy: 2/32 pts above 1.0e−5, worst 4.4e−5, also in the checked mode**
$gg \to e^+e^-\mu^+\mu^-$ 6 legs · QCD · LI-43 OL 2.7e−13 · ML 3.4e−13 · pass · 3 pts	**5.86 s** ×22 validation probes 8.25 s (double-double) 0.69 GiB peak · plan 0.0380 MB Rc 271 ms	**3.75 ms** ×0.70 Rc 3.00 ms · **OL 5.34 ms** · ML 18.0 ms $B = 128$ · 3 runtimes ×1.5 Rc not matched (values excluded): ×1.3 to it <5 ms: not gated on TIRust **TIRust** (tirust) numerator 47 % vs 256-bit truth (32 pts): max 5.4e−12 **NinjaRS per amplitude** (ninjars) 5.31 ms ×0.99 numerator 32 % vs 256-bit truth (32 pts): max 4.9e−10

Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, TIRust and NinjaRS unchecked (re-timed rows)
pyAmpliCol: engine e852f448, native loop-native-v2-slice-e852f448 (c3fda0c4…), engine ecc80899, native loop-native-lit-unchecked-b308254d (0d26af4a…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew, lit-eft) · cells of the 17 re-timed rows: TIRust with the unchecked setting (= auto with it: shown paired on gg_hgg), NinjaRS unchecked beside; the other rows keep their earlier record (auto, checked); gate on TIRust · measured 2026-11-09 · nothing estimated **Protocol:** reduction_unchecked = true (default false) (Runtime.load / evaluate keyword; config evaluation.reduction_unchecked), explicit tirust and ninjars; poles="none", no stability check, no rescue; $B = 128$ per call (median over rounds); one thread pinned to one core (taskset, RAYON_NUM_THREADS=1); the 128 distinct stream points; 3 fresh runtimes per arm in rotated order, the median of their medians (spread ×: largest/smallest); providers F4 most optimistic, timed in the same unit on the same core, alternating with pyAmpliCol; engine ecc80899 (before lit-dress) replication: docs/performance_reports/loop/README.md (unchecked Table 2)		
legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export; optional probes apart (carrier); peak memory; provider generation	median per point at $B = 128$ (fresh runtimes, spread); providers paired (most optimistic); gate or rule on TIRust; stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue (instrumented run: stage shares, not the timed call)
$2 \to 4$		
$gg \to aagg$ 6 legs · QCD · LI-44 Rc 6.8e−6 · OL 6.8e−6 · partial · 3 pts	***not run*** generation beyond the 45-min rule (R9): generate + plan export + exact helicity-zero filter done in 14.0 s (peak 0.63 GiB); the mandatory double-double generation-time probes still running when stopped at 45 min 44 s (2.9 GB RSS; 23:03 UTC); no time **OL 1.54 s** · Rc 1.78 s · ML 3.12 s	
$gg \to zzzz$ 6 legs · QCD · LI-44 Rc 5.2e−6 · partial · 3 pts	***not run*** generation beyond the 45-min rule (R9): generate + plan export + exact helicity-zero filter done in 9.8 s (peak 0.60 GiB); the mandatory double-double generation-time probes still running when stopped at 45 min 45 s (3.7 GB RSS; 23:03 UTC); no time **Rc 5.60 s** · ML 21.4 s · OL unavailable: no OpenLoops ls library serves g g > z z z z (public channel database of OpenLoops 2.1.5, 2026-11-06); row ML/RC only (PLAN §3.2)	
$2 \to 5$		
$gg \to e^+e^-\mu^+\mu^-g$ or $gg \to hgggg$ 7 legs · QCD · LI-51 OL 5.2e−12 · ML 6.6e−12 · pass · 3 pts	**7.62 s** ×6.1 validation probes 88.0 s (double-double) 1.14 GiB peak · plan 0.0530 MB Rc 1.26 s	**71.8 ms** ×0.40 Rc 56.8 ms · **OL 180 ms** · ML 614 ms · $B = 128$ · 3 runtimes ×1.1 Rc not matched (values excluded): ×1.3 to it pass ×0.31 on TIRust **TIRust** (tirust) numerator 60 % vs 256-bit truth (16 pts): max 8.3e−9 **NinjaRS per amplitude** (ninjars) 167 ms ×0.93 NinjaRS 57 % vs 256-bit truth (16 pts): max 9.5e−8
EFT		
rank > N+2 EFT row (HC_NLO_X0 or SMEFT) EFT · LI-X1 · TIRust only · delegated (lit-eft)	***not timed*** execution check only (user 2026-11-06 ~14:55): rank > N+2 has no correct R2 (no dim-8 model), so no reference, no benchmark and no timing claim; lit-eft ran it (two dim-6 insertions through TIRust) no reference provider	



Not drawn: 25 of the 42 rows of Table 2 (3 no pyAmpliCol time at B = 128, 22 no matched provider time). Drawn with a provider that is not matched (its values no reference; no limit from it): $gg \to e^+e^-\mu^+\mu^-$ (Recola), $gg \to e^+e^-\mu^+\mu^-g$ (Recola).

Figure 1: **Batched full colour against the gate** (finite part, $B = 128$; the rows of Table 1 with a matched provider and, as diamonds, the rows of Table 2 with a pyAmpliCol time and a matched provider time).

Table 3: **TIRust and NinjaRS** (the two reduction backends) under the same public unchecked setting: time per point at $B = 128$ with each backend's stage bar (instrumented run), the ratio NinjaRS ÷ TIRust, and their agreement. Loop-induced rows: the re-timed rows of Table 2 (25 other rows not re-timed with the setting at this head), each backend's largest relative error against TIRust at 256 bits (`data/li-unchecked-accuracy.json`) and the binary64 L6 verdict of the checked default (`lit.json`: status, TIRust's served points, the largest finite-part difference of the two backends). Born-interfered rows: Table 1's $2 \to 3$ and $2 \to 4$ rows, with the largest TIRust-vs-NinjaRS difference at the timed points.

Process	TIRust	NinjaRS	N ÷ T	T vs truth	N vs truth	L6 (checked default)
Loop-induced ($\lvert A_{\text{loop}}\rvert^2$) · stage bars: inputs numerator scatter NinjaRS TIR moments contraction masters glue						
$gg \to hh$	**0.437 ms** glue 41 %	**0.612 ms** NinjaRS 34 %	×1.4	3.4e−9	8.8e−11	pass (T 3/3) 3.6e−14
$e^+e^- \to ggg$	**3.42 ms** numerator 56 %	**7.12 ms** NinjaRS 44 %	×2.1	3.5e−7	4.6e−10	partial (T 2/3) 9.6e−8
$gg \to hhh$	**5.20 ms** TIR moments 55 %	**5.65 ms** NinjaRS 48 %	×1.1	7.6e−7	**4.5e64 (2/128 > 1.0e−5)**	pass (T 3/3) 6.6e−13
$gg \to hgg$	**29.2 ms** numerator 55 %	**59.3 ms** NinjaRS 50 %	×2.0	9.2e−10	8.0e−10	pass (T 3/3) 2.7e−10
$gg \to zzg$	**132 ms** numerator 76 %	**386 ms** NinjaRS 64 %	×2.9	**1.2e−2 (18/128 > 1.0e−5)**	5.0e−8	partial (T 2/3) 3.2e−10
$gg \to aag$	**71.1 ms** numerator 78 %	**233 ms** NinjaRS 66 %	×3.3	7.0e−6	**3.1e−3 (1/128 > 1.0e−5)**	partial (T 0/3)
$gg \to w^+w^-g$	**178 ms** numerator 59 %	**540 ms** NinjaRS 70 %	×3.0	**1.4e−2 (24/128 > 1.0e−5)**	6.2e−8	partial (T 0/3)
$gg \to zzz$	**197 ms** numerator 84 %	failed closed: 18/128 failed closed, lane status Unstable	—	2.3e−8	—	pass (T 3/3) 5.3e−11
$gg \to hzg$	**18.4 ms** numerator 76 %	failed closed: 2/128 failed closed, lane status Unstable, lane status MasterFailed	—	8.2e−9	—	pass (T 3/3) 3.8e−13
$gg \to agg$	**115 ms** numerator 70 %	**268 ms** NinjaRS 55 %	×2.3	3.3e−6	**6.8e−4 (1/128 > 1.0e−5)**	partial (T 0/3)
$gg \to zgg$	**191 ms** numerator 65 %	**418 ms** NinjaRS 54 %	×2.2	**3.1e−5 (3/128 > 1.0e−5)**	6.2e−6	partial (T 0/3)

Process	TIRust	NinjaRS	N ÷ T	T vs truth	N vs truth	L6 (checked default)
$u\bar{u} \to hgg$	**1.64 ms** glue 36 %	**3.05 ms** NinjaRS 34 %	×1.9	3.7e−9	3.5e−8	partial (T 2/3) 3.2e−11
$u\bar{u} \to d\bar{d}h$	**0.415 ms** glue 90 %	**0.445 ms** glue 85 %	×1.1	4.1e−10	4.3e−9	pass (T 3/3) 1.3e−10
$gg \to hhgg$	**345 ms** TIR moments 55 %	**789 ms** NinjaRS 75 %	×2.3	7.1e−7	1.8e−6	partial (T 1/3) 3.0e−10; partial (T 0/1)
$gg \to hggg$	**1.44 s** numerator 47 %	**2.53 s** NinjaRS 39 %	×1.8	5.9e−8	**4.4e−5 (2/32 > 1.0e−5)**	partial (T 0/3)
$gg \to e^+e^-\mu^+\mu^-$	**3.75 ms** numerator 47 %	**5.31 ms** numerator 32 %	×1.4	5.4e−12	4.9e−10	pass (T 3/3) 2.6e−13
$gg \to e^+e^-\mu^+\mu^-g$	**71.8 ms** numerator 60 %	**167 ms** NinjaRS 57 %	×2.3	8.3e−9	9.5e−8	pass (T 3/3) 6.9e−12
Born-interfered (Table 1's G1 rows) · LI-B1's L6 over its campaign rows: partial (85 pass, 7 not run, 115 partial, 9 pass (lax)) · `data/backends-born.json` not written yet						
$u\bar{u} \to d\bar{d}g$	not timed: timed in lit-report's final re-time (public API, reduction_unchecked = True)					
$u\bar{u} \to u\bar{u}g$	not timed: timed in lit-report's final re-time (public API, reduction_unchecked = True)					
$u\bar{u} \to e^-e^+g$	not timed: timed in lit-report's final re-time (public API, reduction_unchecked = True)					
$u\bar{u} \to ggg$	not timed: timed in lit-report's final re-time (public API, reduction_unchecked = True)					
$gg \to ggg$	not timed: timed in lit-report's final re-time (public API, reduction_unchecked = True)					
$u\bar{u} \to e^-e^+gg$	not timed: timed in lit-report's final re-time (public API, reduction_unchecked = True)					
$gg \to t\bar{t}gg$	not timed: timed in lit-report's final re-time (public API, reduction_unchecked = True)					
$u\bar{u} \to gggg$	not timed: timed in lit-report's final re-time (public API, reduction_unchecked = True)					
$gg \to gggg$	not timed: timed in lit-report's final re-time (public API, reduction_unchecked = True)					

Missing or incomplete (generated from the tables' data; Table 1 then Table 2; gate lines: time ÷ limit and the stage that dominates pyAmpliCol's time)
T1 · no value: leading colour: not supported (refused by design: EW LC/NLC is never supported (user 2026-10-30, Q4) — $gg \to t\bar{t}e^-e^+g$
T1 · no value: no matched provider time (no gate) — $gg \to gggg$
T2 · no value: providers: not run (no provider timing record of the row in li_campaign.json) — $gg \to h$, $gg \to aa$, $gg \to zz$, $gg \to w^+w^-$, $gg \to za$, $gg \to hz$, $gg \to hg$, $u\bar{u} \to hg$, $ug \to hu$, $gg \to zg$, $gg \to ag$
T2 · no value: no matched provider time (not timed: no production timing yet / unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries) — $h \to gg$, $h \to aa$, $h \to za$, $h \to ggg$, $z \to ggg$
T2 · no value: no matched provider time (not timed: no production timing yet / unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b))) — $aa \to h$, $e^+e^- \to ha$, $aa \to aa$, $aa \to zz$, $aa \to hh$, $e^+e^- \to hh$
T2 · no value: not run: generation beyond the 45-min rule (R9): generate + plan export + exact helicity-zero filter done in 14.0 s (peak 0.63 GiB) — $gg \to aagg$
T2 · no value: not run: generation beyond the 45-min rule (R9): generate + plan export + exact helicity-zero filter done in 9.8 s (peak 0.60 GiB) — $gg \to zzzz$
T2 · no value: not timed: execution check only (user 2026-11-06 ~14:55): rank > N+2 has no correct R2 (no dim-8 model), so no reference, no benchmark and no timing claim — rank > N+2 EFT row (HC_NLO_X0 or SMEFT)
T2 · gate: fail on TIRust — $gg \to hgg$ (×1.2, numerator 55 %), $gg \to agg$ (×1.4, numerator 70 %), $gg \to zgg$ (×1.2, numerator 65 %), $gg \to hhgg$ (×1.5, TIR moments 55 %), $gg \to hggg$ (×2.3, numerator 47 %)
T2 · gate: open on TIRust — $gg \to w^+w^-g$ (×1.4, numerator 59 %)
T2 · validation: partial: L3, L4, L5 (checked default; lit.json) — $h \to ggg$, $z \to ggg$, $gg \to zzz$
T2 · validation: partial: L3, L4, L5, L6; TIRust fails 1 of 3 binary64 fixture points closed (checked default; lit.json) — $e^+e^- \to ggg$, $gg \to zzg$, $u\bar{u} \to hgg$
T2 · validation: partial: L3, L4, L5, L6 (checked default; lit.json) — $gg \to hgg$
T2 · validation: partial: L3, L4, L5, L6; TIRust fails 3 of 3 binary64 fixture points closed (checked default; lit.json) — $gg \to aag$, $gg \to w^+w^-g$, $gg \to agg$, $gg \to zgg$, $gg \to hggg$, $gg \to aagg$, $gg \to zzzz$
T2 · validation: partial: L3, L4, L5, L6; TIRust fails 2 of 3, 1 of 1 binary64 fixture points closed (checked default; lit.json) — $gg \to hhgg$
T2 · accuracy: NinjaRS per amplitude unchecked: 2/128 points above 1e-05, worst 4.5e+64 — $gg \to hhh$
T2 · accuracy: TIRust unchecked: 18/128 points above 1e-05, worst 1.2e-02 — $gg \to zzg$
T2 · accuracy: NinjaRS per amplitude unchecked: 1/128 points above 1e-05, worst 3.1e-03 — $gg \to aag$
T2 · accuracy: TIRust unchecked: 24/128 points above 1e-05, worst 1.4e-02 — $gg \to w^+w^-g$
T2 · accuracy: NinjaRS per amplitude unchecked: 20 points fail (Unstable) — $gg \to zzz$
T2 · accuracy: NinjaRS per amplitude unchecked: 2 points fail (MasterFailed, Unstable) — $gg \to hzg$
T2 · accuracy: NinjaRS per amplitude unchecked: 1/128 points above 1e-05, worst 6.8e-04 — $gg \to agg$
T2 · accuracy: TIRust unchecked: 3/128 points above 1e-05, worst 3.1e-05 — $gg \to zgg$
T2 · accuracy: NinjaRS per amplitude unchecked: 2/32 points above 1e-05, worst 4.4e-05 — $gg \to hggg$

Table 4: **Gate and stage split at $B = 128$**, ms per point: the numbers behind the gate chips and stage bars of Table 1 and behind Figure 1. pyAmpliCol's median (full colour), the binding limit and its provider, the uncertainty band of time ÷ limit, the gate chip of Table 1, and pyAmpliCol's time per stage.

Process	ms/pt	Limit	Band	Gate	inputs	numerator	scatter	NinjaRS	masters	glue
$gg \to gg$	**1.29**	—	—	no gate (2→2)	0.191	0.665	0.0663	0.113	0.0372	0.219
$u\bar{u} \to d\bar{d}g$	**1.72**	1.97 OL	0.862–0.884	<5 ms: not gated ×0.87	0.0768	0.647	0.397	0.142	0.109	0.342
$u\bar{u} \to u\bar{u}g$	**2.64**	3.80 OL	0.678–0.711	<5 ms: not gated ×0.69	0.126	1.10	0.424	0.254	0.190	0.525
$u\bar{u} \to e^-e^+g$	**0.788**	0.948 ML	0.830–0.832	<5 ms: not gated ×0.83	0.0516	0.285	0.129	0.0355	0.0262	0.263
$u\bar{u} \to ggg$	**4.34**	16.1 OL	0.259–0.280	<5 ms: not gated ×0.27	0.389	2.32	0.411	0.456	0.211	0.576
$gg \to ggg$	**20.1**	126 OL	0.155–0.164	pass ×0.16	5.57	9.88	0.518	2.27	0.524	1.14
$u\bar{u} \to e^-e^+gg$	**4.21**	13.3 Rc	0.314–0.319	<5 ms: not gated ×0.32	0.161	2.26	0.589	0.435	0.237	0.524
$gg \to t\bar{t}gg$	**79.8**	613 OL	0.128–0.133	pass ×0.13	11.0	34.9	0.888	18.9	6.63	7.49
$u\bar{u} \to gggg$	**65.3**	486 OL	0.130–0.139	pass ×0.13	9.37	37.2	1.74	9.04	2.98	4.78
$gg \to gggg$	**307**	3 690 OL	0.0824–0.0838	pass ×0.083	93.7	158	2.33	38.4	6.05	8.29
$u\bar{u} \to e^-e^+d\bar{d}g$	**13.9**	45.9 Rc	0.278–0.329	pass ×0.30	0.570	6.62	1.79	1.90	1.27	1.58
$u\bar{u} \to e^-e^+ggg$	**47.9**	326 Rc	0.135–0.159	pass ×0.15	1.56	30.1	2.05	7.93	2.75	3.53
$gg \to t\bar{t}e^-e^+g$	**51.1**	525 OL	0.0897–0.105	pass ×0.097	2.88	29.0	0.734	10.2	4.26	4.07
$gg \to t\bar{t}e^-e^+g$ (EW)	**131**	1 280 Rc	0.0935–0.111	pass ×0.10	6.29	60.7	1.75	29.7	16.5	16.2
$u\bar{u} \to ggggg$	**1 310**	569 000 ML	0.00210–0.00248	pass ×0.0023	215	804	9.27	186	46.0	46.8
$gg \to ggggg$	**4 990**	—	—	no gate (no matched provider)	1 970	2 580	11.0	344	42.0	28.4

Reading the tables

Table 1 (Born-interfered one-loop processes) has one row per test process, by multiplicity. A cell gives pyAmpliCol's value in bold and, at its right, the multiplier *pyAmpliCol ÷ the fastest provider* on this host (in bold among the provider values). A provider time marked ≤ ($u\bar{u} \to ggggg$) comes from a run outside the protocol that does the production work and more (Recola's reference run evaluates the IR poles too), so it bounds that provider's production time from above, and the multiplier against it is a lower bound (≥); it is never a gate limit. Only run times are coloured: green faster, white at parity, red slower ($\log_2$ scale, clamped at ×1/16 and ×16). Generation multipliers are uncoloured (generation time is not the main metric) and leading colour (LC) has no provider ratio. Grey cells say why a value is missing. Run times are medians per phase-space point at $B = 128$, finite part, binary64, one pinned thread, stability checks off; generation is generate + plan export (model loading excluded) with its peak memory and plan size. The providers run in production mode: MadLoop 3.7.0 (ML), Recola 2.2.5 with COLLIER (Rc), OpenLoops 2.1.5 from prebuilt libraries (OL: no generation time). Each cell shows the most recent measured configuration (`scripts/build_summary.py`, by git ancestry of the revisions), named once in the table's first row (engine, native build and digest, date, host); a cell from another configuration names its origin.

Gate chips (Table 1, Figure 1, Table 4): the time over the binding limit, the smallest of MadLoop's time and 1.30 × each of OpenLoops' and Recola's; **pass** or **fail** when the whole uncertainty band (Table 4) lies below or above 1, else **open**. A process below 5 ms per point has no blocking gate ("<5 ms: not gated", user rule of 2026-10-24) and its chip keeps the measured ratio; $2 \to 2$ has no gate. The gate is taken at $B = 128$ only.

Stage bars (every bar of the report, one colour per stage; the cell names the largest share; Table 4 gives every stage of Table 1 in ms): inputs = trees, Born and counterterm inputs (loop-induced: the trees and counterterm kets of the amplitude basis); numerator = seeds, fused factors and the dressing of the open-loop objects; scatter = closures into the reduction lanes; then the reduction backend's own stages: NinjaRS = its expansions (the reduction without the masters) and masters = the master integrals (canonical arguments, cache, oneloopmaster), or TIR moments = TIRust's tensor moments and contraction = their contraction with the numerator (TIRust's master integrals, a part of its tensor moments, are shown apart only where a profiled run recorded them); glue = point parameters, input conversion, transport, squaring, assembly and the rest of the call.

Validation (process cell): the largest deviation $|a - b| \,/\, \max(|a|, |b|, |B|\ \alpha_s/2\pi)$ of the binary64 finite part per reference at the timed points (NJet DD: NJet 3.1.2 in double-double), and the median over the points and references; † the largest lies within pyAmpliCol's binary64 stability estimate at that point; ¶ it is the reference's own: pyAmpliCol's double-double value lies as far from it, and its binary64 value lies within its estimate of the double-double one. ‡ $gg \to ggggg$: the reference check's points 4 and 8 (1.5×10^{-7} and 2.6×10^{-8} from NJet DD) exceed its tolerance 2×10^{-8} in binary64: **accepted (user 2026-10-25): binary64 within the estimate, DD = NJet** (estimates E 1.5×10^{-6} and 3.4×10^{-7}; pyAmpliCol's double-double values equal the reference's); the check keeps its tolerance and, asserting double-double at these points, passes.

Notes under a value qualify it and leave it as measured, in amber a known limitation, in grey its run conditions: $gg \to t\bar{t}e^-e^+g$, full colour ("model: the workspace's py3 G_mu copy"): timed on reference/models/loop_qcd_qed_sm_Gmu_py3 (the harness's processes.json default, as at M-SM), not on the shipped patched UFO (UFOmodeldb 6ca4b120 + the user's issue-1 patch) that the M-SM campaign validated; the speed gate is taken on this timing.

The $2 \to 6$ rows (`summary.json preliminary`, from `data/2to6.json`) have no provider, so no multiplier, gate or colour; their engine is stated once in the table. Generation = generate + plan export, with the store, artifact and load beside it ("FC" marks the

full-colour generation, LC below it); the timed evaluation is binary64, finite part, checks off, one pinned thread, at the largest batch measured. The process cell lists the internal checks (ε^{-2}: double pole against the universal value; E: the stability estimate, with the points above 10^{-5}; rescue: the stream points the default rescue policy's binary64 level sends up; Ward on the external gluons; binary64 against double-double; Catani at full colour; NGluon at leading colour on the all-gluon row) and, in grey, those not run or not asserted.

Table 2 follows Table 1′s rules for loop-induced processes, $|A_{\text{loop}}|^2$ without a Born: one row per process of the M-LIT validation matrix (`m-lit/PLAN.md` §4.1; its ids in the process cell), grouped by multiplicity; the backend-unit, zero, identity, observable and complex-mass-scheme rows (LI-B1, LI-B2, LI-Z1, LI-Z1b, LI-Z2, LI-W1, LI-L1, LI-P1, LI-C1, LI-S1, LI-O1) are validation-only and not listed. On the 17 rows re-timed with the setting (`lit-perf`'s timed rows; the other rows keep their earlier record: the backend `auto` serves, in its checked default), the cell is the batched $B = 128$ time per point of TIRust with the **unchecked reduction setting** (`reduction_unchecked`, a documented public setting, off by default): each backend reduces without its own acceptance or refusal checks. TIRust keeps its a priori conditioning gates and its method choice as method selection, computes no acceptance pass, and serves what the method choice refuses by plain Passarino–Veltman (gate 0); NinjaRS runs with zero stability thresholds; pyAmpliCol's own checks are off. Only what no scheme evaluates finitely (an exactly singular pivot, a non-finite output, a master failure, a NinjaRS lane whose tested quantity is exactly zero or NaN, `Unstable`) still fails. The time is the median over 3 fresh runtimes (created in rotated order) of each one's median over rounds of the public call (`Runtime.load` + `evaluate` on the 128 distinct stream points, poles off, one pinned thread); the spread (×, largest over smallest runtime) is beside B; `auto` with the setting is TIRust, bitwise and without fallback (paired on gg_hgg). Beneath, NinjaRS with the same setting: its time and multiplier where it served every point, else the points it failed with their cause (a call that fails points is timed whole: no served-point time). The coloured bars are the stage shares (the stages of Table 1′s bars, the reduction split into each backend's own stages) of a **separate instrumented run** with the same setting, points and B on the same core, never the timed call. The setting has **no accuracy certificate**: it is a timing and benchmark mode. Beneath each backend's line, its measured accuracy on the row's first stream points (the same public call, binary64 finite part) against TIRust at 256 bits (`data/li-unchecked-accuracy.json`): the largest relative error, or in red the points above 10^{-5} with the worst and how they were served (TIRust: by the gate-0 PV last resort; NinjaRS: values the checked mode refuses, or also in the checked mode) and the points that fail. The providers run in their most optimistic settings (m-lit F4: OpenLoops with its stability system off, MadLoop without its stability test, Recola without dynamic settings, in its release build), in the same unit on the same core as pyAmpliCol, alternating with it between rounds, their time the median over those rounds; the multiplier against the fastest matched provider colours the cell with Table 1′s scale. A provider whose timing record excludes its values as a reference (Recola's on-shell widths against the fixed-width scheme of the others on the rows LI-43 and LI-51) is timed and shown but is no matched provider (its values miss the 1e-5 agreement of `m-lit/PLAN.md` §4.0 L1; §4.1 names MadLoop and OpenLoops as the references of LI-43 and OpenLoops of LI-51): never the fastest matched provider and never a limit, the cell gives the multiplier to it, uncoloured. Generation, uncoloured, is the engine's generation (model loading excluded): generate + the exact helicity-zero filter + plan export, the multiplier formed on it. The generation-time finiteness, Ward and Goldstone probes are validation, off by default (`lit-probe`; the user enables them, in binary64 with a double-double fallback): where a record ran them their time is shown apart beneath the value with its carrier (every record before `lit-probe` ran them in double-double), never in it; the peak memory is over all stages; the provenance line names pyAmpliCol's host and the providers'. Rows of four legs or fewer ($2 \to 1$, $1 \to 2$, $2 \to 2$, $1 \to 3$) are reported without a gate; from $2 \to 3$ upward the gate is Table 1′s (faster than MadLoop and at most 1.30× each matched OpenLoops and Recola time, on TIRust), "<5 ms: not gated" below 5 ms per point, no gate without a matched provider; verdict chips come with the gate tool's records and are TIRust's (the gated backend, "on TIRust"); where TIRust failed points closed the chip says "no verdict … (gap)": its time is no served-point time and how a verdict is given there is not ruled ("no verdict yet" before the gate tool's records, "no provider time yet" while no provider of the row is timed). Grey cells are explicit markers: **pending** (the package that serves the row, named in the cell, has not landed at this head, or the input file that fills the cell does not exist yet), **not run** (a landed package without a record of the row, or a run stopped before its timing, with its measured progress, e.g. a generation beyond the 45-min rule), **failed closed** or **refused** (the record of the cell's request failed points or was refused: no time), and a provider's reason where it has no matched time. Nothing is estimated or extrapolated; each cell shows the newest record of its row. At this head 42 of 42 rows are served: 39 timed, 2 not run (g g > a a g g, g g > z z z z) inputs: `data/li-timing.json` present, `tools/loop_reference/li_campaign.json (provider_timing_records)` present, `data/lit.json` present; the registry against `lit-ref`'s manifest (`tools/loop_reference/li_campaign.json`): checked (39 rows, 42 ids).

Figure 1 places each row's batched full-colour value of Table 1 (dots) against the providers (diamonds, the darkest the fastest) and the gate limit (red bar, the shaded side fails; grey bar: the limit of a process below 5 ms, not gated; dotted line: 5 ms), for the rows with a matched provider. Beneath them, under a rule, the rows of Table 2 with a batched pyAmpliCol time and a matched provider time, pyAmpliCol as a larger diamond in its colour, on the same axis with their providers and, from $2 \to 3$ upward, the gate limit (MadLoop's time and 1.30× each other matched provider's, the smallest); the rows without both are not drawn, counted in the key by reason.

Table 4 lists, for every row of Table 1 timed at $B = 128$, the numbers its gate chip and stage bar summarise: the binding limit and its provider, the uncertainty band of time ÷ limit (the standard error of pyAmpliCol's median over its timing blocks plus the provider's run-to-run half-range), and the time of each stage. Data: `data/summary.json` and `data/summary-li.json`, written by `scripts/build_summary.py` from the campaign records in `data/` (`README.md`).